

VARUNA Project Prototype Technical Guide

Vessel Attribution through Remote-sensing & Unified Navigational Analytics
Smart India Hackathon 2026 — Problem Statement SIH26143

Executive Summary

VARUNA is an end-to-end automated system designed to detect satellite-observed ocean oil spills, backtrack their physical origin using real ocean currents and atmospheric winds, correlate them against real AIS vessel trajectories, and produce a statistically sound, fully explainable vessel attribution dossier. This document details every function, algorithm, decision mechanism, mathematical equation, and data flow.

1. System Pipeline Architecture

The system operates across six sequential modules. Zero fake, synthetic, or mock data is used; every single data object carries an explicit provenance record.

Stage	Module Name	Core Technology / Function	Output Artifact
1	Satellite Ingestion	GDAL Range Requests on STAC (MPC)	Cloud-Optimized GeoTIFF (COG)
2	Dark-Spot Detection	Adaptive Local Thresholding + Risk Scoring	GeoJSON Polygons & Metrics
3	Drift Back-tracking	Lagrangian 5,000 Particle Dispersion + KDE	50%/90% Density Contours
4	AIS Reconstruction	Track Interpolation & Transponder Dark Check	Buffered GeoJSON Lines
5	Vessel Attribution	12-Feature Renormalized Weighting Model	Ranked Candidate Scores
6	Monte Carlo Audit	Bootstrap CI & Common-Mode Resampling	Separation Verdict & CI Bounds

2. Ingestion & Radiometry Processing

2.1 Windowed Satellite Ingestion (ingest_scene)

Located in `varuna_ml/ingest/preprocess.py`. Fetches radar imagery for an Area of Interest (AOI) directly from Microsoft Planetary Computer without downloading full satellite swaths (~2 GB down to ~15 MB). It uses GDAL HTTP Range Requests to pull only the GeoTIFF tiles covering the requested AOI and writes Cloud-Optimized GeoTIFFs (COGs) to MinIO/S3.

2.2 Operator Upload Radiometry Validation (probe_radiometry)

Located in `varuna_ml/ingest/adopt.py`. Probes value distributions of custom GeoTIFF uploads:

- **Linear Sigma0 (Valid):** Values 0.0 to 5.0. Accepted directly for backscatter detection.

- **Decibels dB (Rejected):** Values < 0 dB (e.g. -25 to -5 dB). Detector converts to dB internally; uploading dB imagery causes double-logarithm errors.
- **Uncalibrated DN (Rejected):** Values > 5.0 or 8-bit integers (0-255). Standard images/quicklooks lack calibrated radar physics.

3. Dark-Spot (Oil Slick) Detection Engine

3.1 Linear Backscatter to Decibels (to_db)

Formula: $dB = 10 * \log_{10}(\sigma_{ma0})$. Oil damps short capillary waves, causing strong negative backscatter contrast.

3.2 Hybrid Land Masking (land_mask_from_backscatter & coastline_mask)

Located in `varuna_ml/detect/landmask.py`. Bright land (> -8 dB) is masked alongside Natural Earth 10m vector land polygons. A deliberate **500m land buffer** is grown outward into the water to absorb coastline vector error, protecting against near-shore false positives.

3.3 Adaptive Local Segmentation (segment)

Downsamples the scene by 32x, applies a 9x9 median filter to estimate local sea background intensity (bg), interpolates back to full resolution, and flags dark pixels: $DarkPixel = (db < bg - contrast_db)$ (default 3.0 dB).

3.4 Morphological Cleanup & Risk Scoring

Applies morphological opening (r=2) to remove speckle, closing (r=4) to join slick fragments, and drops regions < 500 px (0.05 km²). Scores look-alike risk using elongation, convexity, contrast, and area, gated by wind speed suitability (4-9 m/s optimal).

Confidence Equation:

$$Confidence = 0.40 * ContrastTerm + 0.35 * (1 - Risk) + 0.15 * WindTerm + 0.10 * SizeTerm$$

4. Backward Lagrangian Drift Simulation

Located in `varuna_ml/drift/backtrack.py`. Backward particle tracking runs 5,000 particles backwards in time (-15 min timesteps) from the slick footprint to calculate a probability surface of where the spill originated.

Particle Velocity Equation:

$$dx/dt = u_{current} + \alpha * R(\theta) * u_{wind} + RandomWalk(Kh)$$

- **alpha (Wind Drift):** Sampled per particle ~ U(0.02, 0.04) [2-4% of wind speed]
- **theta (Ekman Deflection):** Sampled per particle ~ U(0°, 20°) [Right in NH, Left in SH]
- **Kh (Horizontal Diffusivity):** 10 m²/s (Random walk sigma = $\sqrt{2 * Kh * dt}$)

Bilinear Coastal Renormalization: In ocean models, land cells are NaN. Standard samplers return NaN when touching a single land cell, reducing near-shore current to 0 m/s. VARUNA's sampler renormalizes weights strictly over finite (wet) grid nodes, preserving true near-shore velocities.

5. 12-Feature Vessel Attribution Engine

Located in `apps/api/src/modules/attribution/features.ts`. Scores vessel candidates across 12 explainable features.

Key	Feature Name	W	Mathematical Logic & Condition
F1	Spatial Proximity	0.18	$\exp(-d / 8)$ where d is km distance to origin zone.
F2	Temporal Alignment	0.16	Fraction of vessel fixes inside release window.
F3	Track Intersection	0.14	1.0 if track crosses origin zone, else $1/(1 + d/3)$.
F4	Heading Alignment	0.10	$\cos^2(\text{angle_diff})$ vs slick axis. (N/A if elongation < 2.5).
F5	AIS Dark Period	0.10	Transponder silence in window: $\min(1, \text{gap} / 120 \text{ min})$.
F6	Speed Consistency	0.08	Optimal for discharge underway (4 - 14 knots).
F7	Vessel Type Prior	0.07	Tanker=1.0, Cargo=0.7, Fishing/Tug=0.35, Passenger=0.3.
F8	Origin Density	0.05	KDE origin probability mass along track line.
F9	Draught Change	0.04	Reported draught drop across release window (cargo/slops).
F10	Slick Axis Continuity	0.03	Alignment of track bearing with long axis of slick.
F11	Manoeuvre Anomaly	0.03	Average course changes (delta COG) near origin zone.
F12	Prior Incidents	0.02	Confirmed prior spill records for MMSI: $\min(1, \text{count} / 3)$.

5.2 Data Missingness & Renormalization Rule

Features carry three statuses: MEASURED, MISSING, and NOT_APPLICABLE. Weights are renormalised strictly over measured features so candidates with missing AIS features are not penalized artificially:

Score Renormalization Formula:

$$\text{Score} = 100 * \text{Sum}(\text{Normalised_i} * \text{Weight_i}) / \text{Sum}(\text{Weight_measured})$$

Confidence Floor & Tiers: Measured features < 4 -> INSUFFICIENT_EVIDENCE. Score >= 70 -> STRONG, >= 40 -> MODERATE, >= 20 -> WEAK. If drift forcing was missing, origin estimate is degraded and tier is capped at MODERATE.

6. Monte Carlo Verification & Rank Separation

6.1 Single-Vessel Bootstrap Confidence Intervals (bootstrapCi)

Runs 300 Monte Carlo draws perturbing origin zone scaling (+/- 8%) and AIS fix positions to output 5th to 95th percentile confidence bounds per vessel.

6.2 Common-Mode Resampling & Rank Separation (rankSeparation)

Located in apps/api/src/modules/attribution/separation.ts. Standard bootstrap resamples candidates independently, which treats origin uncertainty as separate accidents per vessel. VARUNA uses **common-mode resampling**: it perturbs the origin zone **once per iteration for the entire field**.

Distinguishability Decision Rule:

If Leader Win Share $P(\text{Leader} > \text{Runner-Up}) \geq 90\%$, the leader is **Statistically Distinguishable**. If < 90%, the verdict explicitly states that candidates are **Not Separable** under measurement noise.